Hot and Cold Quark Matter

A bit of historic review

nothing interesting here

- 1930's, Tompson experiment, discovery of the electron
- 1930's, Rutherford experiment, discovery of the nucleus
- 1960's, Friedman, Kendall, and Taylor, proton structure, shot electron at proton, discovered quarks (why we cannot see independent quarks?)
- 1973, Gross, Wilczek, and Politzer, asymptotic freedom, QCD, Nobel prize 2004

QCD pictorial description

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Cold quark matter

- Collins and Perry, 1975, cold quark matter, asymptotic freedom, quark matter at high density
- Alford, Rajagopal, Wilczek, 1998, color superconductivity, quark matter at high density

Hot quark gluon matter

- high enough temperature $> 2 \times 10^{12}$K, proton neutron cannot exist.
- today we use ion collusions to create small droplets 10^{-14} m size big bang matter
- looking for hot gas, but they behave more like liquid.
- Proof: cold atomic experiments. get hold of cold atom and engineer a thin potential trap. turn off the trap. If gas like, then momentum in all direction, but explosion only happens in two directions.

Importance of holographic theories

$\eta/s \simeq 1/4\pi$

Summary & Questions

- elements up to iron made in stars, heavier elements made in supernovae

The Origin of the Heavy Elements

General review

- The origin of the heavy elements in the universe
- Stellar nucleosynthesis and element formation
- Supernova explosions and heavy element production

what's stopping heavier nuclei to form in stars?

- Coulomb repulsion. Need to overcome the Coulomb barrier.
- in stars, heavier elements more centrual, lighter outer, so e.g. carbon + iron can generate energy, but they're distant apart.

one way to work is neutron capture.

Summary & Questions

- r-process nucleosynthesis: rapid neutron capture process, responsible for producing approximately half of the heavy elements beyond iron in the universe. It occurs in environments with high neutron fluxes, such as supernovae and neutron star mergers.
- s-process nucleosynthesis: slow neutron capture process, responsible for producing the other half of the heavy elements beyond iron. It occurs in environments with lower neutron fluxes, such as asymptotic giant branch (AGB) stars.

Nuclear structure & nuclear force

Nuclear structure phenomena – force

analogy: Van der waals force, short range repulsion and long range attraction.

- nuclear force: short range repulsion and long range attraction.
- falls off at large distance.

saturation

- charge density $\rho(r)$ flat for $r < r_0$, falls off for $r > r_0$.
- binding energy per nucleon B/A saturates for $A > 56$.

shell structure

- magic numbers: 2, 8, 20, 28, 50
- attractive mean field
- symmetry $\Rightarrow$ degeneracy, spin-orbit coupling $\Rightarrow$ splitting
- fermi dirac statics
- Mottelson's quantality $\Lambda = \frac{\hbar^2}{Ma^2V_0}$

Odd–even staggering

- Pairing favors correlated nucleon pairs with $J = 0$:

$$P^\dagger = \sum_i C_i \left[a_i^\dagger a_i^\dagger\right]^{J=0}.$$

- Even nuclei gain pairing energy; in odd nuclei, the unpaired nucleon blocks one state.
- For positive binding energies $B(N, Z)$, a three-point indicator is

$$\Delta_n^{(3)}(N,Z) = \frac{(-1)^{N+1}}{2}\left[B(N+1,Z) - 2B(N,Z) + B(N-1,Z)\right].$$

- Odd–even staggering probes pairing, with additional mean-field and shell effects.

shape and hybridization

- by mixing different orbits represented by spherical harmonics, ...
- $0f_{7/2}$ and $1p_{3/2}$ gap size tunes shape of the nucleus.
- Nilsson model: anisotropic harmonic oscillator potential with a spin–orbit term and a quadrupole deformation term.

Shell-model orbit labels; From the Hamiltonian to $0f_{7/2}$

$$H = \frac{\boldsymbol{p}^2}{2m} + V(r) + V_{\mathrm{so}}(r)\,\boldsymbol{l}\cdot\boldsymbol{s}.$$

- For a spherical mean field, single-particle states are labeled by n, l, $s = \frac{1}{2}$, j, and m_j.
- The standard orbit label is nl_j: n counts radial nodes, while $l = 0, 1, 2, 3, 4, \ldots$ corresponds to $s, p, d, f, g, \ldots$.
- Orbital and spin angular momenta couple to $j = l \pm \frac{1}{2}$, with

$$\langle \boldsymbol{l}\cdot\boldsymbol{s}\rangle = \frac{1}{2}\left[j(j+1) - l(l+1) - s(s+1)\right].$$

- The spin–orbit term therefore splits each $l > 0$ orbit into partners with $j = l + \frac{1}{2}$ and $j = l - \frac{1}{2}$.
- $0f_{7/2}$ means $n = 0$, $l = 3$, and $j = \frac{7}{2}$; it has $2j+1 = 8$ magnetic substates.
- Its partner $0f_{5/2}$ has $2j + 1 = 6$ substates, and the harmonic-oscillator major-shell number is $N = 2n + l$.

Summary & Questions

- page 26, $Y_{30} + \alpha Y_{20}$ where's this α come from?
- just pulled out of his head, as example of playing around...

Reactions

Neutron separation energies and lifetimes

The one-neutron separation energy is

$$S_n(A,Z) = [M(A-1,Z) + m_n - M(A,Z)]\,c^2.$$

Nucleus	S_n (MeV)	$T_{1/2}$
^{12}C	18.7207	stable
^{22}C	0.10 ± 0.64	6.2 ± 1.3 ms
^{16}O	15.6639	stable
^{23}O	2.733 ± 0.134	97 ± 8 ms

- Positive S_n means the nucleus is bound against one-neutron emission.
- ^{22}C lies extremely close to the neutron drip line, and its estimated S_n has a large uncertainty.
- The quoted lifetimes for ^{22}C and ^{23}O are their beta-decay half-lives.

Q values for (d,p) reactions

For neutron transfer to the target ground state,

$$\begin{aligned} Q &= [M(A,Z) + M_d - M(A+1,Z) - M_p]\,c^2 \\ &= S_n(A+1,Z) - B_d, \qquad B_d = 2.224566 \text{ MeV}. \end{aligned}$$

- For $^{48}\mathrm{Ca}(d,p)^{49}\mathrm{Ca}_{\mathrm{gs}}$,

  $$Q = 5.146453 - 2.224566 = 2.921887 \text{ MeV}.$$

  The ^{49}Ca ground state has $J^\pi = \frac{3}{2}^-$ and is dominated by a neutron in the $1p_{3/2}$ orbit, so the transfer has $l = 1$.

- For $^{132}\mathrm{Sn}(d,p)^{133}\mathrm{Sn}_{\mathrm{gs}}$,

  $$Q = 2.398655 - 2.224566 = 0.174089 \text{ MeV}.$$

  The ^{133}Sn ground state has $J^\pi = \frac{7}{2}^-$ and is dominated by a neutron in the $1f_{7/2}$ orbit, so the transfer has $l = 3$.

- Both products are single-neutron states outside doubly magic cores, but ^{133}Sn is much more weakly bound and its reaction is only slightly exothermic.

Jets as probes of QGP

Resolution length of the QGP

- L_{res} is the minimum transverse separation between two color charges in a jet for the QGP to resolve them as independent energy-losing sources.
- If $d_\perp < L_{\text{res}}$, the medium sees the charges coherently as one object; if $d_\perp > L_{\text{res}}$, it resolves the branches and quenches them independently.
- A common parametrization is

$$L_{\text{res}} = \frac{R_{\text{res}}}{\pi T},$$

  where T is the local QGP temperature and R_{res} is a dimensionless model parameter, recent ATLAS facors $1 < R_{\text{res}} < 2$.
- $L_{\text{res}} = 0$ gives immediate, fully incoherent resolution, while $L_{\text{res}} \to \infty$ gives fully coherent energy loss.

Moliere scattering

- Similar to Rotherfold.
- larger momentum transfer, smaller scattering angle.
- Soft drop:

Why OO better than $PbPb$?

- Still produce QGP
- Smaller system, less background, easier to study jet quenching.

Summary & Questions

- One of long range plan: see liquid structure of QGP
- Early universe v.s. heavy ion collisions. how to view matter anti matter asymmetry from the lines on ion collisions on QCD phase diagram?
- what the phase diagram looks like if $m_u, m_d, m_s = 0$? the 1st order line between nuclear and CFL is unclear.

Quantum entanglement in nucl and hep



Summary & Questions

- There's bebate on whether Bell's inequality can be violated via collider experiemnts, what's your take on it? People study entanglement anyways.

Nuclear forces

Simplest example

Meson exchange picture: Yukawa potential, $V(r) \sim e^{-m_\pi r}/r$.

$$V(r) \sim g^2 \frac{e^{-mr}}{r}$$

one pion exchange potential:

$$V_{1\pi}(q) = -\Big(\frac{g_{\pi NN}}{2M_N}\Big)^2 (\boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2) \frac{(\boldsymbol{q}\cdot\boldsymbol{\sigma}_1)(\boldsymbol{q}\cdot\boldsymbol{\sigma}_2)}{q^2+m_\pi^2}$$

$$V_{1\pi}(r) = \Big(\frac{g_{\pi NN}}{2M_N}\Big)^2 \frac{m_\pi^2}{12\pi} (\boldsymbol{\tau}_1\cdot\boldsymbol{\tau}_2) \Big[S_{12}(\hat{r}) \Big(1+\frac{3}{m_\pi r}+\frac{3}{m_\pi^2 r^2}\Big) + \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2 \Big] \frac{e^{-m_\pi r}}{r}$$

tensor operator

$$S_{12}(\hat{r}) = 3(\boldsymbol{r}\cdot\boldsymbol{\sigma}_1)(\boldsymbol{r}\cdot\boldsymbol{\sigma}_2) - \boldsymbol{\sigma}_1\cdot\boldsymbol{\sigma}_2$$

Off-shell NN potential

idea of something being off-shell: $E^2 \neq p^2 + m^2$.

From QCD to chiral symmetry

For the light-quark doublet $q = (u, d)^T$,

$$\begin{aligned}\mathcal{L}_{\text{QCD}} &= -\frac{1}{4} G^a_{\mu\nu} G^{a\mu\nu} \\ &\quad + \bar{q}\left(i\gamma^\mu D_\mu - \mathcal{M}_q\right) q, \\ \mathcal{M}_q &= \text{diag}(m_u, m_d).\end{aligned}$$

- In the chiral limit $\mathcal{M}_q \to 0$, define

  $$q_{L,R} = P_{L,R} q, \qquad P_{L,R} = \frac{1 \mp \gamma_5}{2}.$$

  The quark Lagrangian separates into left- and right-handed sectors and is invariant under

  $$q_L \to L q_L, \quad q_R \to R q_R,$$
  $(L, R) \in SU(2)_L \times SU(2)_R$.

- Finite quark masses couple the two chiralities,

  $$\mathcal{L}_m = -\bar{q}_L \mathcal{M}_q q_R - \bar{q}_R \mathcal{M}_q q_L,$$

  and explicitly break chiral symmetry.

- The QCD vacuum also breaks the symmetry spontaneously:

  $$SU(2)_L \times SU(2)_R \longrightarrow SU(2)_V, \qquad \langle \bar{q} q \rangle \neq 0.$$

- The three pions are the resulting pseudo-Goldstone bosons; their small masses obey, at leading order,

  $$m_\pi^2 f_\pi^2 = -(m_u + m_d)\langle \bar{q} q \rangle + \cdots .$$

Chiral EFT (χEFT)

- At momenta $Q \sim \{p, m_\pi\} \ll \Lambda_\chi \sim 1$ GeV, the active degrees of freedom are nucleons and pions; heavier QCD physics is absorbed into low-energy constants.
- The most general Lagrangian consistent with QCD symmetries is organized as

$$\mathcal{L}_{\chi\mathrm{EFT}} = \sum_\nu \mathcal{L}^{(\nu)}, \qquad \mathcal{L}^{(\nu)} \sim \left(\frac{Q}{\Lambda_\chi}\right)^\nu .$$

- For an irreducible two-nucleon diagram, Weinberg power counting gives

$$\nu = 2L + \sum_i \Delta_i, \qquad \Delta_i = d_i + \frac{n_i}{2} - 2,$$

  where L is the number of loops, d_i counts derivatives or powers of m_π, and n_i counts nucleon fields at vertex i.
- The nuclear interaction is expanded as

$$V = V_{1\pi} + V_{2\pi} + V_{\mathrm{contact}} + V_{3N} + \cdots .$$

  At LO (Q^0), one-pion exchange and nonderivative contacts appear; derivative contacts and two-pion exchange enter at higher orders, while the leading three-nucleon force appears at N^{2}LO (Q^3).
- The truncated potential is iterated in the Lippmann–Schwinger equation,

$$T = V + V G_0 T.$$

  Observables are order-by-order predictions, with truncation errors controlled by powers of Q/Λ_χ.

BSM and symmetries and EFT

new physics why

neutrino mass, dark matter, baryon asymmetry, strong CP problem, hierarchy problem, inflation, dark energy.

new physics how

- colliders, probing high energy
- precision measurements, probing low energy (across AMO, HEP, NP)

one important way is rare or SM-forbidden processes. (L, B, CP, L_α) , $0\nu\beta\beta$ decay, proton decay, n-nbar, EDMs, LFV.
Another way is precision tests of SM-allowed processes: beta decay, PV electron scattering, muon g-2, EDMs, LFV, LHC precision measurements.
searches/characterization of light and weakly coupled particles.

Sakharov conditions for baryogenesis (1967)

Observed baryon asymmetry (from CMB and BBN):

$$\eta = \frac{n_B}{n_\gamma} \approx 6 \times 10^{-10}.$$

To dynamically generate $\eta \neq 0$ from a symmetric initial state, Sakharov identified three necessary conditions:

- **Baryon number (B) violation.** Without it, any reaction creating baryons is balanced by its reverse; $n_B - n_{\bar{B}}$ is conserved and cannot grow from zero.
- **C and CP violation.** C symmetry alone equates the rates of $X \to \bar{q}\bar{q}$ and $\bar{X} \to qq$. CP violation is needed so that baryons and antibaryons are produced at genuinely different rates.
- **Departure from thermal equilibrium.** In equilibrium, CPT symmetry enforces $\Gamma(X \to f) = \Gamma(\bar{X} \to \bar{f})$, so any asymmetry is immediately washed out. A non-equilibrium epoch — e.g. the cosmic expansion or a first-order phase transition — is required to "freeze in" the asymmetry.

The Standard Model has all three ingredients in principle:

- B violation via electroweak *sphalerons* (non-perturbative transitions active at $T \gtrsim T_{\text{EW}} \sim 100$ GeV).
- CP violation through the CKM phase δ_{CKM}.
- Out-of-equilibrium dynamics at the electroweak (EW) phase transition.

Yet the SM *fails* to explain the observed asymmetry:

- The effective SM CP violation is suppressed by many small Yukawa couplings: $\delta_{\text{CP}}^{\text{SM}} \sim 10^{-20} \ll \eta \sim 10^{-10}$.
- For $m_H \approx 125$ GeV the EW transition is a smooth *crossover* (not a first-order PT), providing no sharp out-of-equilibrium epoch.

$\Rightarrow$ BSM physics is required to explain the matter–antimatter asymmetry of the universe.

probing UV new physics via EFT

Key point (Appelquist–Carazzone decoupling, 1975): a heavy particle of mass M affects low-energy physics at $E \ll M$ only through

- a *shift* in the coupling constants of existing interactions, and
- *new local interactions* suppressed by powers of E/M, consistent with the underlying symmetries.

Analogy: second-order perturbation theory in QM. Split $H = H_0 + \lambda V$, with unperturbed eigenstates $H_0|n^{(0)}\rangle = E_n^{(0)}|n^{(0)}\rangle$. The perturbed energy to order λ^2 is

$$E_n(\lambda) = E_n^{(0)} + \lambda\langle n^{(0)}|V|n^{(0)}\rangle + \lambda^2 \sum_{k\neq n} \frac{\langle n^{(0)}|V|k^{(0)}\rangle\langle k^{(0)}|V|n^{(0)}\rangle}{E_n^{(0)} - E_k^{(0)}} + \mathcal{O}(\lambda^3).$$

For states $|k^{(0)}\rangle$ with $E_k^{(0)} \gg E_n^{(0)}$ ("heavy" states), the second-order term can be absorbed into an *effective Hamiltonian* acting only in the low-energy sector:

$$\langle n^{(0)}|H_{\text{eff}}|n^{(0)}\rangle + \mathcal{O}\left(\frac{E_n^{(0)}}{E_k^{(0)}}\right), \qquad H_{\text{eff}} = \lambda^2 \sum_{k\neq n} \frac{V|k^{(0)}\rangle\langle k^{(0)}|V}{-E_k^{(0)}}.$$

The heavy states enter only through the *energy denominators* $1/E_k^{(0)}$, suppressing their effect by powers of E/M.
The EFT translation:

- Low-energy eigenstates $|n^{(0)}\rangle \leftrightarrow$ light fields at $E \ll M$.
- Heavy eigenstates $|k^{(0)}\rangle \leftrightarrow$ heavy particles of mass M being *integrated out*.
- $H_{\text{eff}} \leftrightarrow$ effective Lagrangian: new operators of dimension $d > 4$ suppressed by $1/M^{d-4}$.

$\Rightarrow$ You do not need to know the UV completion to make precise low-energy predictions; its effects are encoded in a finite set of Wilson coefficients multiplying low-energy operators.

SMEFT: Standard Model Effective Field Theory

What it is. SMEFT is the EFT built on top of the SM itself. Take the SM as the *renormalizable* (dimension-4) low-energy theory and systematically add all higher-dimension operators consistent with the SM gauge symmetry $SU(3)_C \times SU(2)_L \times U(1)_Y$ and field content:

$$\mathcal{L}_{\text{SMEFT}} = \mathcal{L}_{\text{SM}} + \sum_{d=5}^{\infty} \sum_{i} \frac{C_i^{(d)}}{\Lambda^{d-4}} \mathcal{O}_i^{(d)}.$$

Λ is the (unknown) scale of new physics being integrated out; $C_i^{(d)}$ are dimensionless *Wilson coefficients* encoding the UV model details. **Operator counting.**

- **dim-5** (Weinberg operator, 1979): unique up to flavour, $\mathcal{O}_5 = (LH)(LH)$. After EW symmetry breaking it gives Majorana neutrino masses $m_\nu \sim v^2/\Lambda$.
- **dim-6**: 59 independent operator classes (Warsaw basis, Grzadkowski et al. 2010) after field redefinitions; these are the workhorse of SMEFT phenomenology.
- dim-7, dim-8, ...: suppressed by additional powers of E/Λ; important only for specific observables (e.g. neutrinoless double-beta decay at dim-7/9).

What problems does it tackle?

- **Model-independent BSM searches.** Rather than testing one specific BSM model at a time, SMEFT simultaneously parameterises *all* heavy new physics through the Wilson coefficients. Experimental constraints on C_i/Λ^2 apply to entire classes of UV theories.
- **Higgs and EW precision physics.** Dim-6 operators modify Higgs couplings, triple gauge vertices, and oblique parameters. LHC global fits constrain $\sim$50 operators simultaneously.
- **Flavour and CP violation.** Four-fermion dim-6 operators mediate FCNC processes (K–$\bar{K}$, B physics), probing $\Lambda \sim 10^3$–10^5 TeV.
- **Neutrino masses.** The dim-5 Weinberg operator is the lowest-order explanation; its coefficient is directly tied to the seesaw scale.
- **Connecting UV to IR.** Given a specific BSM model (e.g. heavy Z', leptoquark, seesaw mediator), matching at the scale Λ gives definite predictions for all C_i, which can then be mapped to observables — a clean top-down workflow.

Key limitation. SMEFT assumes $E \ll \Lambda$ throughout; if the new physics is light or strongly coupled, a different EFT (or the full model) is needed.

Neutron star and neutron star merger

Historic review

discovery of first pulsar, 1967, Jocelyn Bell Burnell and Antony Hewish, Nobel prize 1974.

Optical pulsation of crab

bright flash followed by a dimmer flash, 33 ms period. Rotates extremely fast.

$$\omega^2 R < GM/R^2 \Rightarrow \rho > \frac{3}{4\pi}\frac{\omega^2}{G}$$

Typical numbers: $T = 33$ ms, $R = 10$ km, then $\rho > 1.2 \times 10^{11}$ g/cm^3

Electron capture at high densities

- Electron density above 10^{11} g/cm^3: $\rho = \rho_e m_n / Y_e$.
- Electron fraction Y_e is electrons per baryon, ~ 0.5, nucleon mass m_n.
- Electron Fermi momentum k_F: $\rho_e = k_F^3/3\pi^2$. $k_F > 0.1$ fm$^{-1} = 20$ MeV (units $\hbar c = 1 = 197.3$ MeV fm)
- Large Fermi energy $E_F \sim k_F = 20$ MeV $[(Y_e/0.5)(\rho/10^{11}$ g/cm$^3)]^{1/3}$ drives electron capture $e + p \to n + \nu$.
- Rotational period of 33 ms $\Rightarrow$ Crab is both very dense and neutron rich.

TOV equation

Hydrostatic equilibrium for a spherical star in general relativity.

$$\frac{dP}{dr} = -\frac{G(\varepsilon + P)\left(M + 4\pi r^3 P\right)}{r^2\left(1 - 2GM/r\right)}, \qquad \frac{dM}{dr} = 4\pi r^2 \varepsilon$$

- $\varepsilon(r)$ energy density, $P(r)$ pressure, $M(r)$ mass enclosed within r.
- Newtonian limit $dP/dr = -GM\varepsilon/r^2$ recovered when $P \ll \varepsilon$ and $2GM/r \ll 1$.
- Closed by an equation of state $P(\varepsilon)$: integrate out from $r = 0$ ($M = 0$, $P = P_c$) to $P = 0$, giving R and $M(R)$.

SM symmetries & SMEFT overview